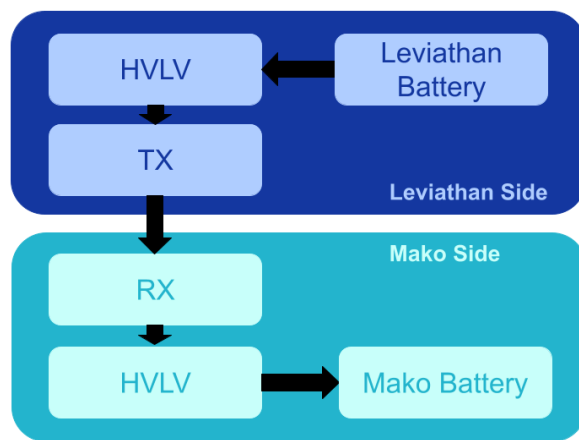
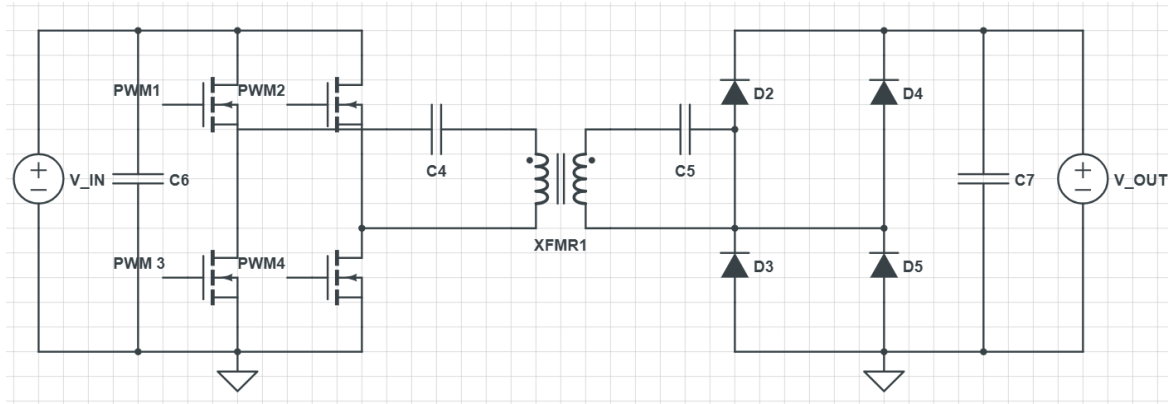


ULYSSES LARS WIRELESS CHARGER

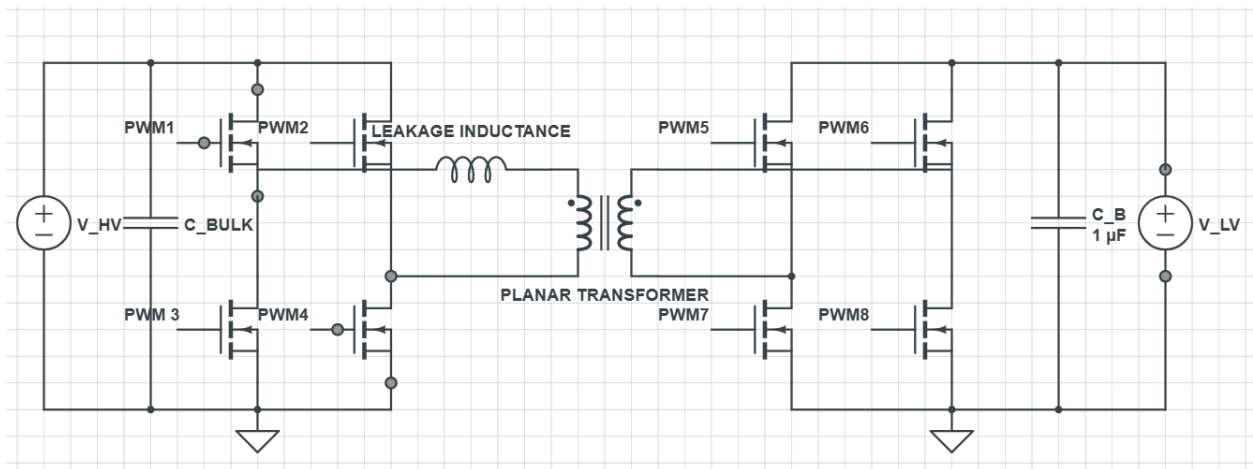
The charging system of the Mako 2.0 aims to achieve a maximum of 3kW max power transfer rate at 400VDC input. This can charge an 8kWh battery in 2.6 hours, ideal for returning the Mako to operation as quickly as possible. This requires a power interface between the Mako and a charging dock aboard the Leviathan. This technical brief describes the hardware and firmware topologies for the design, which were all developed in-house.



The charging system is a magnetic resonant wireless charger, utilizing a topology standard in EV wireless charging research. It utilizes three main electrical systems: a bi-directional dual active bridge acting as a power converter between low and high voltage, a high-voltage H-bridge, and a high-voltage rectifier.



Schematic diagram of RCTX



Schematic diagram of HVLV

Operations of Wireless Charging

Magnetic resonance wireless charging takes advantage of Faraday's law of induction: that a moving magnetic field generates a moving electric field. The wireless component comes from the fact that a magnetic field does not need to be in contact with a material in order to induce a changing voltage, thus power can be "transferred" across an air gap. A changing magnetic field can be generated using one or more switches connected in series with a DC voltage. The maximum flux is generated using an H-Bridge. Total magnetic flux is a function of voltage magnitude and frequency, so charging systems can transfer higher power at higher voltages and frequencies. The relationship between power, frequency, and voltage in wireless charging systems is roughly linear.

When a coil drives a sinusoidal input voltage into another coil, their magnetic fields mutually induce sinusoidal waveforms; this is mutual inductance. The field coupling of two coils is a nonlinear function of distance, angle in 3D space, and the magnetic permeability of the material between the coils (typically air). The total field coupling can be described by coupling coefficient k , where k lies between 0 and 1. For traditional transformer topologies, k is typically closer to 1. For wireless topologies, k is typically closer to 0.

Driving frequency is one of the most important parameters to tune. Higher frequency means higher power transferred because the magnetic flux increases linearly with input frequency. Higher frequency also means higher switching losses, which are quite significant at high power transfer. If not careful, large enough switching losses can destroy the switching elements. In high power wireless chargers, the driving frequency lies in the hundreds of kHz range, between 80kHz and 120kHz, with the standard being 85kHz. This range is chosen as a delicate balance between maximizing power transfer and minimizing switching losses.

The resonance of the system is the most important target to hit. When capacitors are put in series or parallel with the coils, they form an LC resonant tank with the coils. This LC resonant tank creates gain, which increases the field coupling. The benefits are maximized when the transmitter and receiver coils have the same inductance and capacitance. There are a number of compensation topologies which increase in complexity, the conventional method is to use a single series capacitance.

Resonance becomes more complicated with drifting in inductance and capacitance due to frequency or manufacturing variation. The resonant frequency can change during operation. Driving the system away from resonance can significantly decrease the power transferred, depending on the quality factor of the LC tank, so the system needs to constantly drive the LC tank at its resonant frequency, no matter what it is.

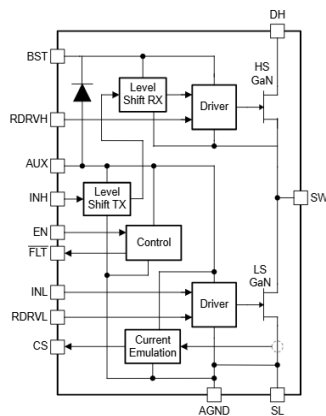
The receiver needs to convert high frequency AC from the coil into DC. This is typically accomplished with an H-bridge using high power Schottky diodes, since the system needs to switch at high frequency. Diodes have a particularly high switching and conduction loss compared to transistors, so receivers need to have careful thermal management. At the output of the diode H-bridge, a bulk capacitance smooths the output voltage to create steady DC. The necessary bulk capacitance decreases with higher driving frequency, which is another reason to push the driving frequency as high as possible.

TX Hardware

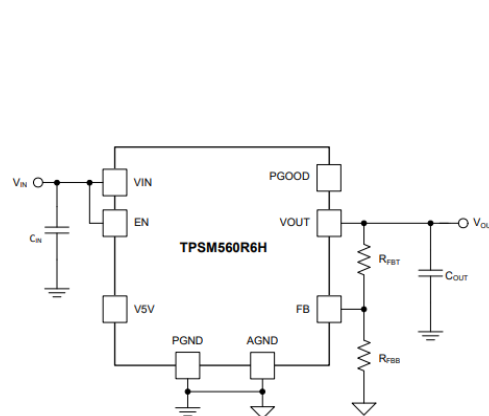
The high-voltage H-bridge acts as a driver for a transmitter coil. It is designed to switch a high-voltage square wave between 80kHz and 120kHz. Due to the high switching losses at this switching speed, the TX system uses GaNFETs instead of MOSFETs. The main benefit of GaN is improved thermal efficiency at high switching speed. The drawback of GaN is that FETs have a high slew rate, which causes problems with ringing from parasitics. The gate drive and switching loop layout for GaNFETs need to be carefully packaged to minimize parasitic inductance.

The TX system uses the LMG2610 from Texas Instruments, which packages the FETs and gate drives into a single chip to minimize the effects of poor layout on switching performance. The LMG2610 is rated for a 600V bus and drain current of 6.4A. From testing, the LMG2610 is seen to have a low current draw while driving, reducing the size of the auxiliary gate drive converters. The LMG2610 also acts as the star ground point between power ground and analog ground. The drive converter is based on the TPSM540 from Texas Instruments, a small buck converter with an integrated inductor. The TPSM540 drives the gate drive power rails and the digital power rail.

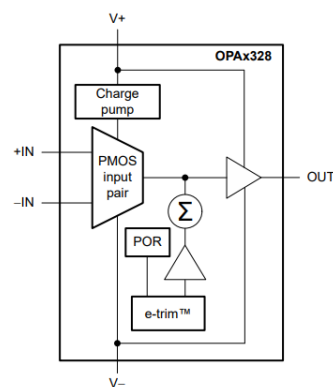
The LMG2610 has a current emulation pin that outputs a scaled version of the drain current, that can be read by an ADC through a current sense resistor. This voltage is fed into the OPA2328, a low noise op-amp that outputs a scaled version of the waveform for the ADC. The on-board compute for the TX is the STM32H723ZGT6, which has advanced hardware timers for PWM generation and a high-speed ADC.



LMG2610



TPSM560

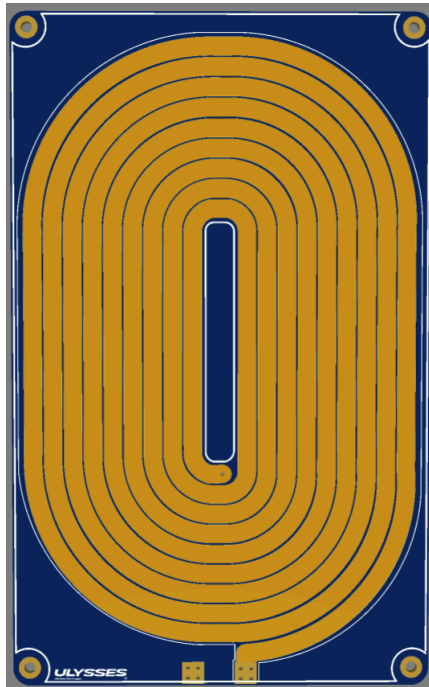


OPA2328

RX Hardware

The high-voltage diode bridge acts as a rectifier, outputting steady DC voltage. The total bulk capacitance is 11 μ F, which for a load of 6A at 100kHz, produces a voltage ripple of 0.136%. The diode bridge consists of 12 total MURSDA860 schottky diodes, paralleled in each bridge to increase the current handling capability and to increase the thermal area. The current through the LC tank is read through a TMCS1133 hall effect sensor, an isolated current sensor that outputs a small signal waveform. The RX computing module is the STM32G431, which has a high-speed ADC in a small package.

PCB Coils



Wireless charging coils are typically made of Litz wire, wrapped in a spiral formation. The RXTX instead uses coils etched onto a PCB, referencing a research paper. The original paper experiments with an etched coil for an induction stove, targeting 20-30A of current through the coil, with a trace width of 10mm. The PCB coils used in RXTX have a width of 5mm, sized appropriately for the current going through the coil.

The main benefits of PCB coils are consistent inductance, improved packaging, ease of manufacturing, and control over shape. Unlike litz coils, PCB coils do not require a dedicated manufacturing house or in-house manufacturing tools. They have a consistent inductance, series resistance, and quality factor that do not drift significantly due to manufacturing variance. This makes resonance tracking and impedance matching significantly easier, improving the system performance by a large margin. They offer improved packaging due to mounting holes and a dielectric surface. They are easier to tune due to their ease of design.

The main drawback of the PCB coil is poor thermals, they increase in temperature significantly more than litz wire coils. This requires a cooling mechanism on the bottom of the coil, connected to the ferrite backing. The total inductance of the PCB coils is 12uH, a series resistance of 0.8 ohms, and a quality factor of 12.

RXTX Firmware

RXTX is entirely based on digital controls on the STM32H723ZGT6. This chip was chosen due to its high-speed ADC, programmable dead time, and advanced timers TIM1 and TIM8, allowing for ease of application on power converters. The primary goal for the controls is to track the resonant frequency of the system at any given time, and being able to adapt the driving frequency to the changing resonant frequency. This requires an ADC input from the H-bridge to track power through the H-bridge, feeding into a control algorithm. The controls are based on a hill-climbing algorithm, using current and frequency readings as inputs and outputs.

There are two stages to the controls: static, and dynamic resonance tracking. Static resonance tracking sweeps across the total frequency band with preset resolution and boundaries, marking each point with a current reading. The algorithm finds the frequency with the highest current reading, and finishes by driving the system at resonance. Dynamic resonance reads the current at a consistent interval, and perturbs the system in the direction of highest descent, like a gradient-descent algorithm.

The ADC output is inherently difficult to work with because it is a changing waveform. From the GaNFET H-Bridge, the output is approximately a sine wave with positive offset. Another challenge is the noise from the ADC. The ADC readings are inherently noisy due to EMI from RXTX and from the harmonics due to the oscillation from the current sense output, so there are high frequency spikes in the waveform. With these factors in mind, it is easy for the algorithm to form the wrong conclusions from the raw ADC data. The firmware filters the data and generates an aggregate rms current reading, eliminating high frequency harmonics and voltage spikes. The filtering algorithm needs to be fast enough to perform in real time on thousands of data points. The filtering algorithm uses C's standard library QSort, and eliminates the highest few elements that stand out as outliers, which has a time complexity of $n \log(n)$.

During bench testing, it was found that static resonance tracking usually finds a frequency close but not exactly at resonance, and dynamic resonance tracking eventually moves the system to resonance. This confirms the validity of the static and dynamic control architecture.

HVLV Hardware

The system requires high voltage to perform at high efficiency at 3kW. To step-up the Leviathan battery voltage and to step it down for the Mako battery voltage, the system uses a bi-directional dual-active bridge, a common topology for this kind of power conversion.

The DAB uses a planar transformer to reduce the BOM cost, as high-power transformers can be as expensive as the rest of the BOM in its totality. The DAB uses an STM32H723ZGT6 for digital controls, taking advantage of its advanced hardware timers TIM1 and TIM8 for phase-shifting between bridges.

The high voltage bridge uses the IPB60R120 MOSFETs, a high-voltage, high-current FET. The low voltage bridge uses the BSC190N, and parallels two FETs per branch of the bridge to increase the thermal dissipation area and reduce conductive and switching losses per switch.

The DAB uses the UCC1414 for gate drive power isolation, the ISO7740 for PWM isolation, the ISO224 for analog signal isolation, and the UCC33420 for digital power isolation. Gate drive and digital power rails are generated by the TPSM560. The gate drive for both the high voltage and low voltage bridges are driven by the UCC27741 high voltage gate driver for low and high side gate drive, with a rating of 650V.

